

Md Salauddin Sultan

Senior SDET | Test Automation | Backend & Database Testing | Java, Python, Shell | Security Validation
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Detail-oriented Software Development Engineer in Test (SDET) with 10 years of experience, specializing in Database Testing, PL/SQL development, and Web Application Testing & Automation. Proven expertise in validating complex database systems, ensuring data integrity, and testing end-to-end web-based applications.

Strong experience in **Oracle databases**, **backend validation**, and **automation** using Java, Python, and Shell scripting. Adept at working in Agile environments, improving test coverage, and delivering high-quality, scalable, and secure applications.

Work Experience

Oracle Corporation, Bengaluru, India

- Principal Member of Technical Staff Oct-2025 – April-2026
 - **Led** test automation initiatives, mentoring team members and collaborating with cross-functional stakeholders to drive quality and efficiency in Agile environments.
 - **Partnered** with engineering managers, product owners, and business stakeholders to assess risks, define testing priorities, and ensure delivery objectives.
 - Led **Common Criteria (CC) certification for Oracle Database 26ai**, driving security validation efforts and ensuring compliance with international security standards.
 - Demonstrated **extensive experience** in PL/SQL development, query optimization, and backend data validation across multiple Oracle database versions.
 - Identified and reported **150+ security** defects and **250+ functional** issues, significantly enhancing product stability and security posture.
 - **Mentored** junior engineers in automation scripting, testing best practices, and version control, improving team productivity and code quality.
 - Explored AI-assisted testing approaches using LLM to improve test design, automation development, and root-cause analysis.
 - Delivered **presentations** on Unified Auditing and DBSAT to familiarize the team with new features and capabilities.
- Senior Member of Technical Staff Dec-2020 – Oct-2025
 - **Designed, executed and Automated** database-centric test strategies for Oracle products, including Unified Auditing, Database Security Assessment Tool (DBSAT), and Privilege Analysis.
 - Contributed to the **stabilization** of the unified audit feature during a critical phase driven by escalations from top customers. As part of the test team, expanded cross-functional and end-to-end test coverage, uncovering 350+ defects and accelerating system stabilization within a few months.
 - Leveraged strong **expertise** in PL/SQL, Java, Python, and scripting languages to validate complex backend systems and workflows.
 - **Designed and maintained** scalable automation frameworks using Python and PyTest, supporting regression, end-to-end, and CI/CD integrated testing, improving efficiency and scalability.
 - Designed and optimized **CI/CD pipelines**, reducing deployment time and minimizing manual intervention.
 - Defined and tracked automation KPIs including regression coverage, defect leakage, execution time, and release readiness metrics.
 - Increased regression test coverage from **70% to 90%**, reducing risks and improving overall product quality.
 - Proactively enhanced **cross-functional test runs**, improving the stability and reliability of the Unified Auditing feature.
 - Took end-to-end **ownership of feature validation**, from requirement analysis to production readiness, ensuring quality deliverables.
 - **Collaborated** with developers and project managers to define acceptance criteria, test strategies, and quality benchmarks.
 - Built internal **QA tools** such as Regress Finder, Diff Checker, and TDBaaS for regression detection, log comparison, and automated test data/database setup.
 - **Owned and maintained** high-quality regression suites integrated with daily build pipelines, ensuring early defect detection and improving release confidence.
 - Partnered closely with development teams to debug backend and database-related issues.
- Member of Technical Staff Feb-2018 – Dec-2020
 - Played a key role in **Common Criteria (CC) certification testing for Oracle Database 19c**, validating security controls and ensuring adherence to global security standards.

- **Strengthened** cross-functional testing for the unified auditing area by expanding test scenarios and collaboration across teams, leading to earlier defect detection and improved release quality.
- **Validated** data integrity, stored procedures, triggers, and complex SQL workflows to ensure system reliability and accuracy.
- Performed **end-to-end testing** of web-based applications, ensuring seamless integration between UI, API, and database layers.

CGI Inc., Bengaluru, India

Nov-2015 – Feb-2018

- Associate Software Engineer
 - **Developed** a web-based Sentiment Analysis application with interactive visualizations (bar charts, pie charts, word clouds), enabling effective analysis of user feedback from social media platforms.
 - **Developed an automated** hourly email notification system to report job execution status, improving team visibility, and reducing manual monitoring.
 - **Built utilities** such as Job Report and Data Upload tools to automate reporting workflows and streamline data processing.

Core Skills

- Database & Backend: Oracle DB (11g-26ai), PL/SQL, SQL Query Optimization, Data Validation
- Programming & Automation: Java, Python, Playwright, Selenium, Perl, Shell Scripting, PyTest, C++, Generative AI, Prompt Engineering, AI-assisted Automation.
- Web Technologies: JavaScript, REST APIs, Web Application Testing.
- Testing & Automation: Test Automation, CI/CD, API Testing (Postman).
- Core CS: Data Structures, Algorithms.
- Tools & Platforms: SQL Developer, IntelliJ, Eclipse, Postman, Visual Studio, Jira, Confluence.
- Operating Systems: Linux/Unix, macOS, Windows.

Key Expertise

- Test Automation Framework Development
- API & Backend Testing
- Database Testing & Data Validation
- Web Application Testing
- CI/CD & DevOps Integration
- Scalable Test Execution using Containers (Docker, Kubernetes)
- Data Integrity & Security Testing

Certification

- Understanding of Agentic AI certified Professional.
- Oracle Cloud Infrastructure Certified Generative AI Professional.
- Foundation: Building Reliable Agents.
- Generative AI: Introduction to Large Language Models.
- Qualified GATE 2011, 2012 and 2014.

Education

R. V. College of Engineering | Bengaluru | India

2012-2014

Master of Technology

Software Engineering | 71.5%

Cochin University of Science and Technology | Kochi | India

2007-2011

Bachelor of Technology

Computer Science and Engineering | 64.43%